

Resolution Geometry of the Wall–Tap Experiment

The Whole Theorem Ladder on its Origin: from Under-Determination to Saturation, from Local Charts to Walls, and Closure at Scale

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Abstract

The wall–tap observation is the origin of Resolution Geometry (RG). Here it is declared as a family of acoustic protocols and taken through the whole theorem ladder [1]. The latent parameters are source position x , wall contrast r and tap rhythm b . The nuisances are gain g , sound speed c and room length L , and the environment is wall matching ε , wind w and the noise family.

The record is under-determined at every stage until the protocol grows. Infinitesimal nullity falls 5, 4, 3, 2, 2, 1, 0 by kernel intersection as delay, echo, amplitude, overlap and lead/lag channels are added, and then the length and gain anchors.

The overlap channel hides an exact $\mathbb{Z}_2$ fibre $b \leftrightarrow 2d(x) - b$: 56% of the parameter domain has a partner with identical law. The regular quotient is therefore $(x, r, |b - d|)$, not (x, r, b) as previously stated, and the fold $b = d(x)$ is an isotropy wall inside the regular stratum. Only a new signed channel saturates it. At the matched wall the laws specialize under Hellinger continuity, with Fisher rank $3 \rightarrow 1$ and Cramér–Rao bound $O(\varepsilon^{-2})$, while a shrinking-covariance family fails the gate.

Distance needs anchors. Delays fix only ratios. A rod, or thermometry together with a counted clock, anchors length. A temperature profile is a blind sector, and wind turns echo distance into the acoustic Landau–Lifshitz metric.

Closure is attacked at scale: 3000 random requests across seven modules, five protocols, walls, folds, anchors, wind and noise families, dispatched with real certificates. The result is 0 false LOCKED on every seed, all seven statuses, and false LOCKs from each of seven sabotaged certificates. One of these, a scale-free rank test, revealed that certificates must be stated in declared precision units.

The battery passes 16/16 checks on five seeds and eight canaries are caught.

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1 Claim discipline

The historical observation motivates a declared family of protocols: a triggered tap, delayed returns, echo magnitudes, rhythmic overlap, and comparison across sessions. The law below is synthetic. It is not a reconstruction of a room, and nothing in it identifies a latent room, filter or connection. Every claim names its rung of the ladder and its registered check (WT01–WT16). Rungs that this experiment cannot instantiate are listed with the status RG assigns to them (Section 11).

2 The declared experiment

Definition 2.1 (Wall-tap observation system). • *Latent parameters:* $\theta = (x, r, b) \in K = [0.7, 4.3] \times [0.25, 1] \times [0.004, 0.028]$.

- *Nuisance:* gain g , sound speed c and room length L (declared values 1, 343.2 m/s and 5 m).
- *Environment:* wall contrast $\varepsilon \geq 0$, wind w , and the noise family.

With $d(x) = (2L - 2x)/c$, the protocols are cumulative:

$$\begin{aligned}
P_0 : t_{\text{dir}} &= x/c, & P_1 : +t_{\text{echo}} &= (2L - x)/c, & P_2 : +a &= \varepsilon g r e^{-0.08(2L-x)}, \\
P_3 : +o &= \varepsilon g r e^{-((b-d)/w_0)^2}, & P_4 : +s &= \varepsilon g r \tanh((b-d)/w_0),
\end{aligned}$$

with $w_0 = 3.5$ ms. The noise is Gaussian with fixed diagonal covariance (standard deviations 25 and 40 μ s; 0.035, 0.025, 0.025). The anchors are:

- A_L : a rod of length L ;
- A_c : thermometry, $c = \sqrt{\gamma k_B T/m}$, together with a counted sample clock;
- A_g : a reference reflector of known contrast;
- A_w : a reciprocity (two-way transducer) channel.

3 From under-determination to saturation

Theorem 3.1 (Saturation by kernel intersection; L1–L3, L35; WT01). *In (x, r, b, g, c, L) the Fisher rank of the joint law is 1, 2, 3, 4, 4 for $P_0, \dots, P_4$. It becomes full (4 of 4) with A_c , A_L and A_g , and also full (5 of 5) with A_L and A_g alone, since two delays then fix c . Each step is an intersection of kernels, so nullity never increases.*

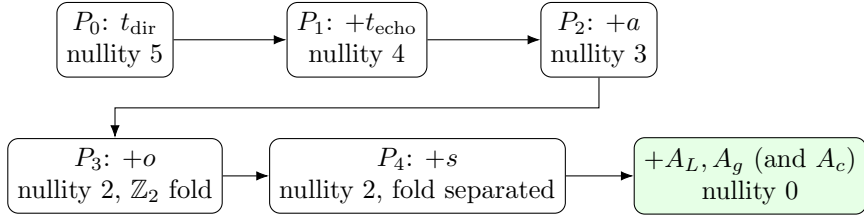


Figure 1: From under-determination to saturation. Each protocol intersects the kernel (L3). Continuous nullity in (x, r, b, g, c, L) falls until the anchors close it. The fold is discrete, so it is removed by the signed channel and not by rank.

Proof. The kernel of a joint law is the intersection of the protocol kernels (L3); adding channels can only shrink it. The ranks are computed at 30 points per protocol, away from the fold. $\square$

Proposition 3.2 (The exact $\mathbb{Z}_2$ fibre; L1, L2, L10; WT02). *For P_3 and $\varepsilon > 0$, the laws at (x, r, b) and $(x, r, 2d(x) - b)$ are identical (difference 0). The partner lies in K for 55.5–55.9% of K across seeds. Hence:*

- the exact quotient of P_3 is $(x, r, |b - d(x)|)$;
- P_3 has full differential rank in (x, r, b) off the fold but does not separate the pairs, which is L10 (rank, separation and completeness are distinct) realized by the origin experiment;
- the reflection is a discrete gauge element (L6).

The signed lead/lag channel of P_4 separates every pair by more than one noise unit. Saturation needs a new protocol, not more data.

4 Two walls

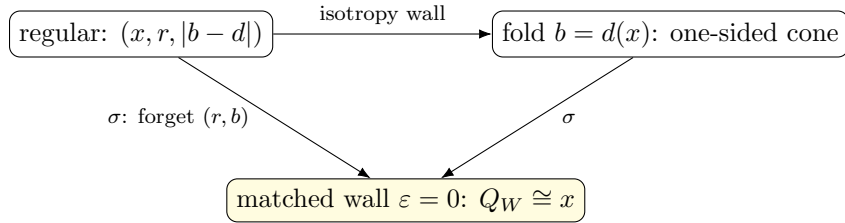


Figure 2: Strata of the wall-tap quotient. Arrows are non-invertible specializations (L18). Regular transport is invertible only within each stratum (L21).

Proposition 4.1 (Fold wall; L15, L16; WT03). *At $b = d(x)$ the b -derivative of P_3 vanishes (value 0), so the fold is a latent rank wall inside the regular stratum. Opposite displacements in b change the overlap identically. The jet-rescaled secant $\Delta o/h^2$ converges $(-48979.6, \text{convergence } 4 \times 10^{-6})$ to a one-sided direction that the first-order tangent misses. Off the fold the rank is 3.*

Proposition 4.2 (Matched-wall specialization; L17–L21; WT04). *For $\varepsilon \rightarrow 0$:*

- limits of regular kernels lie in the wall kernel span (∂_r, ∂_b) (defect $\leq 10^{-9}$ over 200 points);
- $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$: fold pairs and contrast pairs are identified at the wall;
- the specialization map $[x, r, |b - d|] \mapsto [x]$ exists and is many-to-one.

5 Distinguishability at the matched wall

Theorem 5.1 (Hellinger, Fisher and decision specialization; L22, L30; WT05–WT07). *With fixed covariance:*

- $\sup_K H(F_\varepsilon, F_0) = 0.9999, 0.653, 0.185, 0.047$ for $\varepsilon = 0.2, 0.05, 0.0125, 0.003$, with slope 0.98 (analytically $O(\varepsilon)$);
- the efficient (r, b) -information given x scales as ε^2 (slope 2.000);
- the Cramér–Rao bound grows by two decades per decade of ε ;
- the wall Fisher has rank 1 with radical exactly $\text{span}(\partial_r, \partial_b)$;
- the minimal error for testing $r = 0.3$ against $r = 0.8$ is 0.027, 0.423 and 0.492 for $\varepsilon = 0.5, 0.05, 0.005$, tending to $1/2$.

No-Go 5.2 (Shrinking covariance; WT05, WT07). With $\Sigma_\varepsilon = \varepsilon^2 \Sigma$ on the reflected channels, the means converge but $\sup_K H = 1.0$ at every ε , and the test error stays 5.6×10^{-5} . The H-hell gate rejects this family. A gate based on means alone accepts it (canary).

6 Nuisance, natural descent and statistical distance

Proposition 6.1 (Gain fibre; L27; WT08). *Echo amplitude depends on gr. Without a reference reflector the efficient information on r is 0 (fraction 0.0). With one it is restored (fraction ≥ 0.57). A certificate that silently fixes g (the non-efficient metric) locks r falsely (WT16).*

Proposition 6.2 (Natural descent and experiment relativity; L13, L39, L43; WT12). *The Fisher tensor is invariant under linear charts of the (whitened) record, with exact cocycle (error 9×10^{-11}). RG distance scales with $\sqrt{N}$ in the number of taps (factor 10.000 for $N = 1 \rightarrow 100$), while displacement ratios in x are invariant (ratio 3.000), by translation covariance of the delay channels.*

7 Scale, units, media and wind

Theorem 7.1 (Dilation and anchors; L70–L73, L78; WT09). *Two delays in (x, c, L) have rank 2, with the exact dilation $(x, c, L) \mapsto (kx, kc, kL)$ as null direction. They fix x/L and L/c only. Either of two anchors fixes length:*

- a rod (L) , which gives rank 2 in (x, c) ;
- thermometry, $c = \sqrt{\gamma k_B T / m} = 343.22 \text{ m/s}$ at 293.15 K, together with a counted clock, which gives rank 2 in (x, L) .

Thermodynamics alone fixes a velocity, not a length (Buckingham, L70); the counted clock supplies the time unit.

Proposition 7.2 (Media; L82; WT10). *Opposite linear temperature gradients and a uniform profile with the same harmonic mean give identical delays (8.6657 ms), while the endpoint sound speeds differ by 6 m/s. The profile is a blind sector until path diversity or local thermometry is added (H-med).*

Proposition 7.3 (Wind; L79–L81; WT11). *In wind w along the path, echo distance is $D/(1 - w^2/c^2)$: 3.0001 and 3.0016 m at $w = 2$ and 8 m/s for $D = 3 \text{ m}$. This is the acoustic Landau–Lifshitz metric of a stationary, non-static medium. A reciprocity channel recovers w exactly from $\frac{D}{2}(t_{AB}^{-1} - t_{BA}^{-1})$. In still air, echo and rod distance agree.*

8 Cone and clocks

Proposition 8.1 (Acoustic hyperbolic lock; L46, L52; WT13). *The symbol $(\tau + w\xi)^2 - c^2\xi^2$ has real roots for every w (imaginary part 0). Still or moving air gives a convex, tilted characteristic cone. An anti-damped symbol $\tau^2 + c^2\xi^2$ has roots growing with frequency (rate $c = 343.2$ per unit frequency): Hadamard ill-posed, and outside the physical domain.*

Proposition 8.2 (Clock groupoid; L61–L63; WT14). *The sample clock and the metronome share a kernel, and their transition is affine (residual 0). A temperature-sensitive clock has a different kernel, and no transition map exists (non-functional by 8.8 ms).*

9 Latent transport

Proposition 9.1 (No law-natural room transport; L58–L60; WT15). *Within the contrast-gain fibre, a vertical shear preserves the law (change 1.2×10^{-6} , the finite-difference accuracy) and maps one lift onto another exactly. Two declared priors give room transports that differ by 45.9° in the (r, g) plane. How the room “moves” is declared, not observed.*

10 Closure at scale

Theorem 10.1 (Closure on the wall–tap domain; L87–L90; WT16). *Seven modules were requested:*

- the metric on (x, r, b) ;
- global rhythm b ;
- distance x in metres;
- the wall metric on x ;
- contrast at the wall;
- a quantum metric;
- the latent room.

They were drawn at random over protocols, wall contrasts, near-fold rhythms, gain reference, anchors, wind, reciprocity, environment declaration and noise family. With real certificates (double-escalation rank in declared precision units, the fold test, the H-hell gate, anchor and wind gates), the 3000 requests per seed gave:

seed	locks	false locks	power	statuses reached
20261004	267	0	1.000	7/7
1	269	0	0.993	7/7
42	262	0	1.000	7/7
777	279	0	0.996	7/7
2026	269	0	1.000	7/7

Status counts on the primary seed:

CORRESPONDENCE 1275, OUTSIDE DOMAIN 951, LOCKED 267, ESCALATE 256, VACUOUS 122, STRUCTURAL WALL 93, INCONSISTENT 36.

Seven sabotaged certificates were run through the same dispatcher. False LOCKs per seed, over 1500 requests:

sabotage	rung violated	false LOCKs (five seeds)
numerical rank of noisy data	L38	9–21
scale-free (normalized) rank test	L38, L27	11–19
ignore the overlap fold	L10, L16	18–25
means-only wall gate (weak convergence)	L30	3–10
silent sound speed	L70, L82	49–76
silent gain (non-efficient metric)	L27	3–8
ignore wind	L80, L81	44–58

Remark 10.2 (Certificates in declared units). The first closure run gave 7 false LOCKs. The rank certificate used a correlation-normalized Fisher matrix. Near the matched wall the entire (r, b) block collapses uniformly as ε^2 , so the normalized matrix stays well conditioned, and a scale-free certificate is blind to the wall. Certificates must state the efficient information in declared precision units. This sharpens L38 and L88: the gate error α_g is defined only relative to a declared target precision.

11 Ladder coverage

rungs	content	wall-tap status
L0–L5	observation system, kernel, quotient, joint kernel, garbling	instantiated: WT01, WT02
L6–L11	symmetry, gauge, saturation hierarchy, coherence	instantiated: fold gauge, L10 (WT02)
L12–L16	tangent sequence, natural descent, strata, tangent cones	instantiated: WT03, WT12
L17–L21	specialization across walls	instantiated: WT04
L22–L30	Fisher, efficient metric, nuisance, data processing, H-hell	instantiated: WT05–WT08
L31–L34	quantum metric module	OUTSIDE DOMAIN (classical acoustic law); the dispatcher refuses (WT16)
L35–L38	persistent null, nulls, finite-sample rank	instantiated: WT01, WT16
L39–L43	RG distance, experiment relativity	instantiated: WT12
L44–L53	order, cone, residual branch, hyperbolic lock	instantiated: WT13; L48 (Lorentz, $n \geq 2$) not instantiable in 1D
L54–L60	scale cocycle, transport, connection	instantiated: WT15; L54–L55 trivial (no curvature of scale)
L61–L64	clocks, duration	instantiated: WT14
L65–L69	dynamics, entropy, chaos cones	not instantiated (periodic tapping); a dynamical tapping protocol must be declared, otherwise OUTSIDE DOMAIN
L70–L84	units, anchors, media, stationary slices, physical distance	instantiated: WT09–WT11; L76–L77 (multi-dimensional rigidity) and L83 (ergodic transfer) not instantiated
L85–L90	kernel–stabilizer assembly, global transport, closure	instantiated: WT15, WT16

rungs	content	wall–tap status
L91–L92	external and extensions	not applicable

12 Corrections to the earlier wall–tap global paper

1. The regular quotient is $(x, r, |b - d|)$, not (x, r, b) . The overlap channel identifies b with $2d - b$ on 56% of K (Proposition 3.2).
2. The fold $b = d(x)$ is a wall inside the regular stratum, with a one-sided tangent cone. The earlier “away from the overlap symmetry locus” restricted the rank statement but not the quotient.
3. “LOCKED on regular strata” holds for the local metric away from the fold. Global rhythm is CORRESPONDENCE under P_3 and LOCKED only under P_4 .
4. The declared sound speed 343.2 m/s is a medium speed, not an SI constant. Physical length needs a rod or thermometry with a counted clock (H-med), and wind changes echo distance.
5. The rung numbers of the earlier figures (L16.x, L24.1) are replaced by the final numbering.

The earlier paper’s correct results are retained: Hellinger specialization with fixed covariance, the shrinking-covariance canary, the Fisher collapse, chart invariance, and the held-out calibration principle.

13 Non-claims

- The law is a declared synthetic feature law, not a model of a particular room.
- Fixed covariance is part of the protocol.
- Finite sampling does not certify uniform convergence for arbitrary waveforms; H-hell is proved analytically here.
- Nothing identifies a unique latent room, filter or connection.

A Reproducibility

`rg_wall_tap_attack.py` passes 16/16 checks on seeds 20261004, 1, 42, 777 and 2026. `--canary` runs eight sabotages: the means-only Hellinger gate and the seven certificates of Theorem 10.1. All are caught.

Corrections made during development are recorded in the code:

1. WT06 first took eigenvalues of the raw Fisher matrix, whose scales span 10^{15} ; it now uses the efficient block.
2. WT09 first registered rank 1 for two delays in (x, c, L) ; the rank is 2, with one dilation null.
3. WT12 first transformed raw features; charts now act on whitened features.
4. WT15 first assumed $JB = I$ for an over-determined record; lifts now satisfy $JB = P_{\text{im}J}$.
5. WT16 first used a normalized rank certificate, which gave 7 false LOCKs. This was the finding of the remark on declared units.
6. WT16’s first gain sabotage was absorbed by the efficient certificate; it was replaced by the non-efficient metric.

References

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